



Boolean Algebra

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Boolean Operations & Expressions

- *Variable* – a symbol used to represent a logical quantity.
- *Complement* – the inverse of a variable and is indicated by a bar over the variable.
- *Literal* – a variable or the complement of a variable.

Basic Boolean Equations

- For the basic gates/functions
- AND
 - $Z = A B$
 - $X = C D E$ 3 input gate
 - $Y = F G H K$ 4 input gate
- OR
 - $Z = A + B$
 - $Y = F + G + H + K$ 4 input gate
- NOT
- $F = \bar{Z}$

Boolean Algebra theorems

- 1. $X + 0 = X$
- 2. $X + 1 = 1$
- 3. $X \cdot 0 = 0$
- 4. $X \cdot 1 = X$
- 5. $X + X = X$
- 6. $X \cdot X = X$
- 7. $X + \bar{X} = 1$
- 8. $X \cdot \bar{X} = 0$
- 9. $\bar{\bar{X}} = X$

Boolean Algebra theorems

10. $X + Y = Y + X$

11. $X \cdot Y = Y \cdot X$

12. $X + (Y + Z) = (X + Y) + Z$

13. $X(YZ) = (XY)Z$

14. $X(Y + Z) = XY + XZ$

15. $X + XY = X$

16. $X + \bar{X}Y = X + Y$

17. $X + YZ = (X + Y)(X + Z)$

18. $X + \bar{X}Y = X$

DeMorgan's Theorems

- The complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables.
- The complement of two or more ORed variables is equivalent to the AND of the complements of the individual variables.

$$\overline{X \bullet Y} = \bar{X} + \bar{Y}$$

NAND Negative-OR

$$\overline{X + Y} = \bar{X} \bullet \bar{Y}$$

NOR Negative-AND

DeMorgan's Theorems

- Very important in simplifying equations
 - $\overline{(X + Y)} = \bar{X} \cdot \bar{Y}$
 - $\overline{(XY)} = \bar{X} + \bar{Y}$

X	Y	X+Y	$\overline{X+Y}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

X	Y	$\bar{X}$	$\bar{Y}$	$\bar{X} \cdot \bar{Y}$
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

Complement of a function

- In real implementation sometimes the complement of a function is needed.
 - Have $F = \bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z$

$$F = \bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z$$

$$\bar{F} = \overline{\bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z}$$

$$= \overline{(\bar{X}Y\bar{Z})} \cdot \overline{(\bar{X}\bar{Y}Z)}$$

$$= (X + \bar{Y} + Z) \cdot (X + Y + \bar{Z})$$

DeMorgan's Theorems (Exercises)

- Apply DeMorgan's theorems to the expressions:

$$\overline{X \bullet Y \bullet Z}$$

$$\overline{X + Y + Z}$$

$$\overline{\overline{X} + \overline{Y} + \overline{Z}}$$

$$\overline{\overline{W} \bullet \overline{X} \bullet \overline{Y} \bullet \overline{Z}}$$

DeMorgan's Theorems (Exercises)

- Apply DeMorgan's theorems to the expressions:

$$\overline{(A + B + C)D}$$

$$\overline{ABC + DEF}$$

$$\overline{A\overline{B} + \overline{C}D + EF}$$

$$\overline{\overline{A + B\overline{C}} + D(\overline{E + \overline{\overline{F}}})}$$

Prove the following expression by using truth table

- $Y = XYZ + \bar{X}Y + XY\bar{Z}$

X	Y	Z	$\bar{X}$	$\bar{Z}$	XYZ	$\bar{X}Y$	$XY\bar{Z}$	$XYZ+\bar{X}Y$	$XYZ+\bar{X}Y+XY\bar{Z}$
0	0	0	1	1	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	1	1	0	1	0	1	1
0	1	1	1	0	0	1	0	1	1
1	0	0	0	1	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0
1	1	0	0	1	0	0	1	0	1
1	1	1	0	0	1	0	0	1	1

Prove the following expression by using truth table

- $(X+Y)(X+\bar{Y})=X$

X	Y	$\bar{Y}$	$(X+Y)$	$(X+\bar{Y})$	$(X+Y)(X+\bar{Y})$
0	0	1	0	1	0
1	0	1	1	1	1
0	1	0	1	0	0
1	1	0	1	1	1

Simplify

- Simplify logical expression to its simplest form.
 - $F = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ$

$$F = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ$$

$$= \bar{X}Y(Z + \bar{Z}) + XZ \quad \text{by identity 14}$$

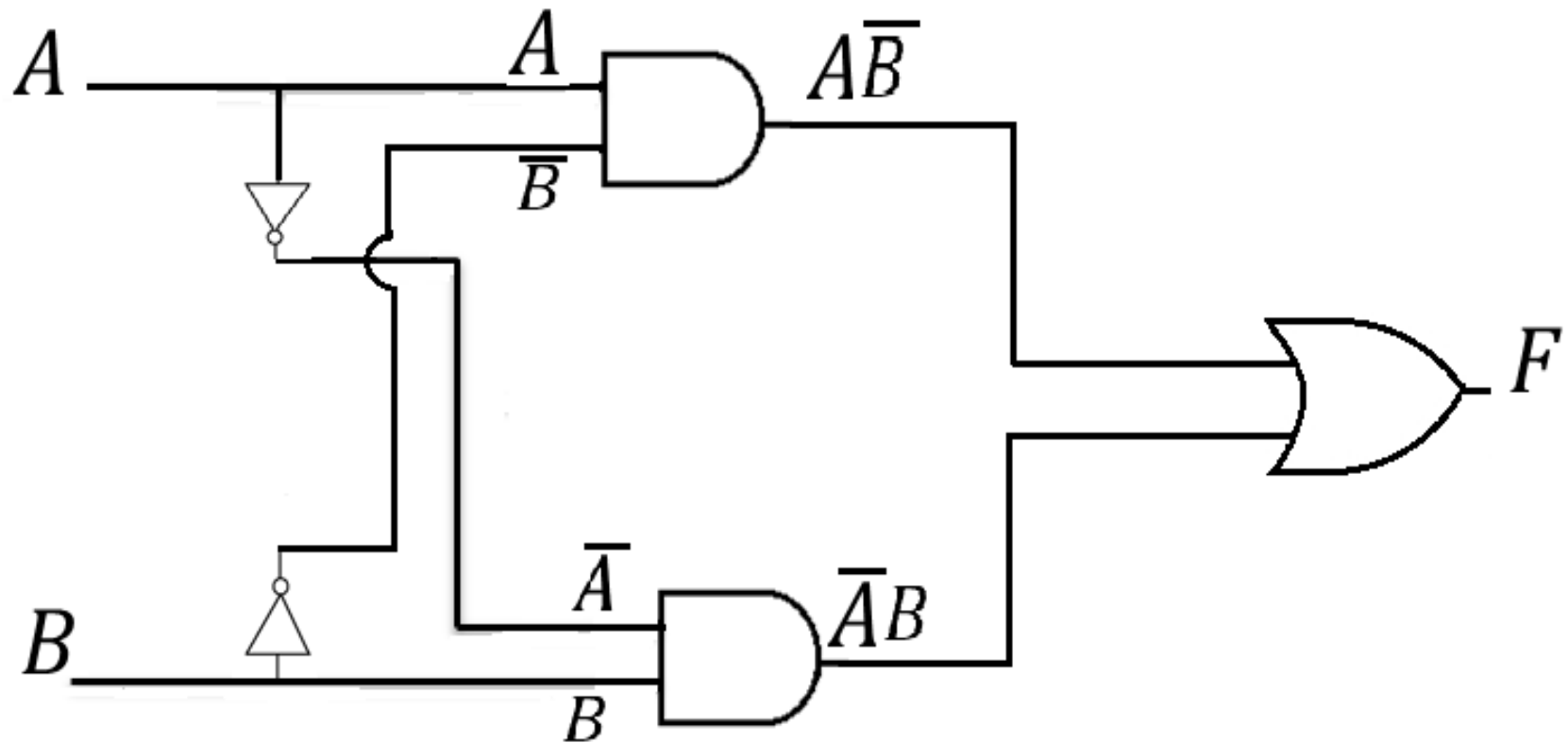
$$= \bar{X}Y \cdot 1 + XZ \quad \text{by identity 7}$$

$$= \bar{X}Y + XZ \quad \text{by identity 2}$$

Simplify

- Draw the logic diagram for the following expression :

$$F = A \bar{B} + \bar{A} B$$



Simplify the following expression

$$F = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ$$

$$F = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ$$

$$= \bar{X}Y(Z + \bar{Z}) + XZ \quad \text{by identity 14}$$

$$= \bar{X}Y \cdot 1 + XZ \quad \text{by identity 7}$$

$$= \bar{X}Y + XZ \quad \text{by identity 2}$$

Simplify the following expression and draw a logic diagram for the simplified expression

$$F = \bar{A}BC + A\bar{B}C + ABC + B\bar{C}$$

SOLUTION

$$F = \bar{A}BC + ABC + A\bar{B}C + ABC + B\bar{C} \quad \text{theorem (5)}$$

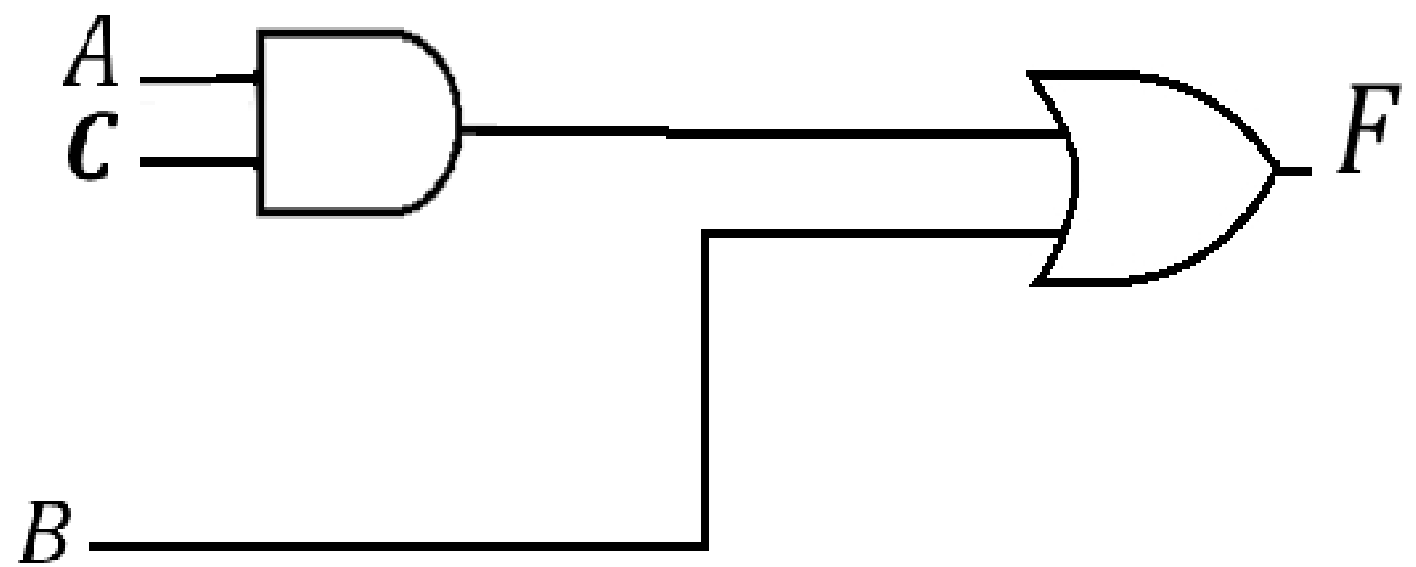
$$F = (\bar{A} + A)BC + A(\bar{B}C + BC) + B\bar{C}$$

$$= (1)BC + A(\bar{B} + B)C + B\bar{C}$$

$$= BC + AC + B\bar{C}$$

$$= B(C + \bar{C}) + AC$$

$$F = B + AC$$



Simplify the following expression and draw a logic diagram for the simplified expression

$$F = Y W \bar{Z} + YWZ + XY$$

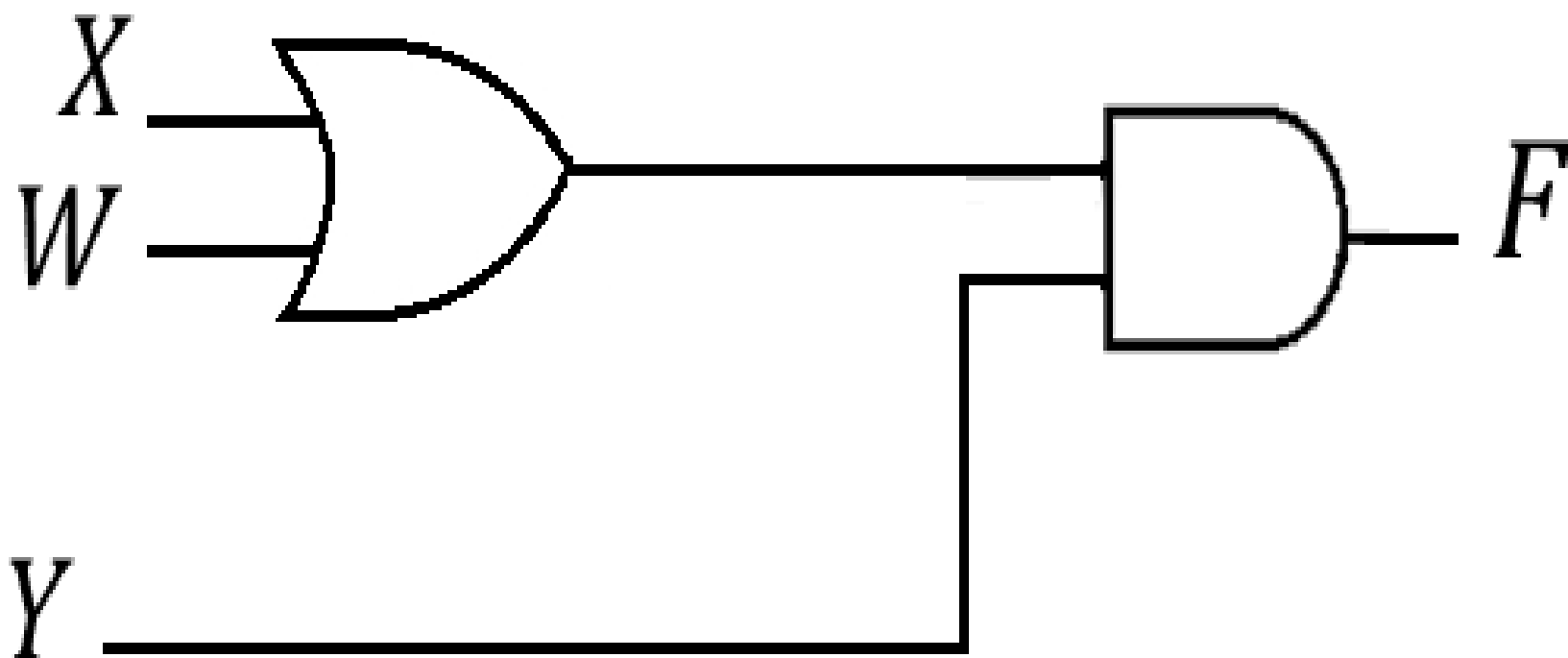
- SOLUTION

$$F = Y W \bar{Z} + YWZ + XY$$

$$F = (Y W) (\bar{Z} + Z) + XY$$

$$F = (Y W) (1) + XY$$

$$F = Y (W + X)$$



Simplify the following expression and draw a logic diagram for the simplified expression

$$F = A B C + A B \bar{C} + A \bar{B} C + \bar{A} B C$$

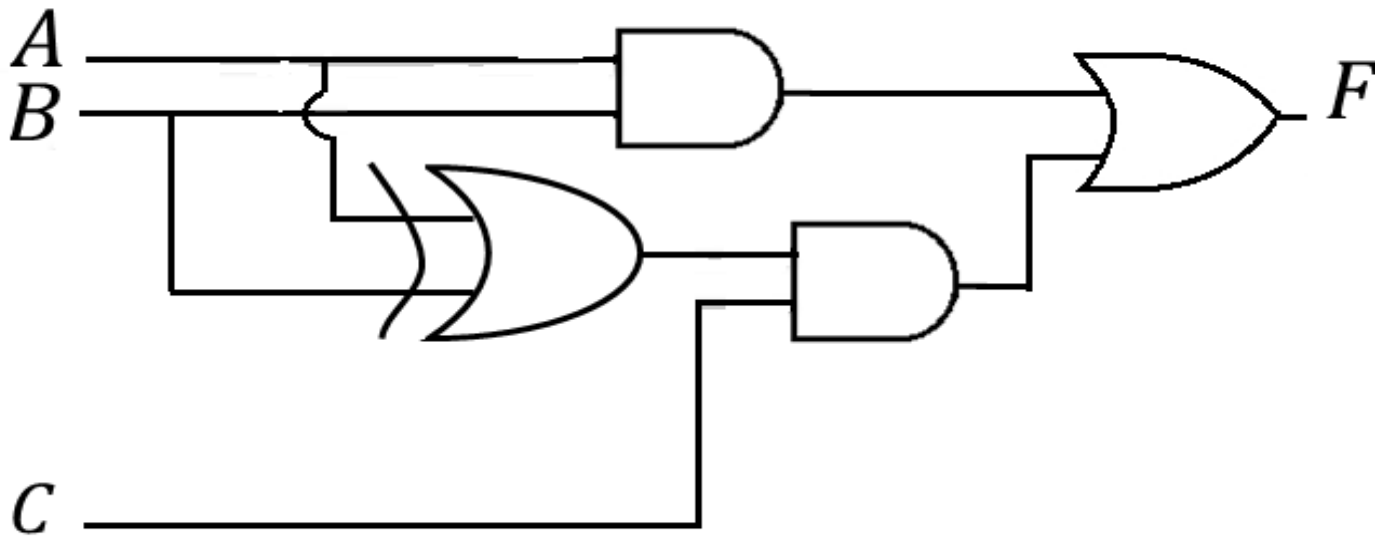
- SOLUTION

- $F = (A B (C + \bar{C}) + A \bar{B} C) + \bar{A} B C$
- $= (A B + A \bar{B} C) + \bar{A} B C$
- $= A B + (A \bar{B} C + \bar{A} B C)$
- $= A B + (A \bar{B} + \bar{A} B) C$
- $= A B + (A \oplus B) C$

Simplify the following expression and draw a logic diagram for the simplified expression

$$F = A B C + A B \bar{C} + A \bar{B} C + \bar{A} B C$$

- SOLUTION
- $F = A B + (A \oplus B) C$



Simplify the following expression and draw a logic diagram for the simplified expression

$$F = A B(DA + C\bar{D}) + \bar{A}\bar{B}(A + C)$$

SOLUTION

- $F = A BDA + ABC\bar{D} + \bar{A}\bar{B}A + \bar{A}\bar{B}C$
- $= A BDA + ABC\bar{D} + \bar{A}\bar{B}A + \bar{A}\bar{B}C$
- $= A B(\textcolor{red}{D} + \textcolor{red}{C}\bar{D}) + 0 + \bar{A}\bar{B}C$ identity 16
- $= A B(\textcolor{red}{D} + \textcolor{red}{C}) + \bar{A}\bar{B}C$
- $= A BD + ABC + \bar{A}\bar{B}C$
- $= (A B + \bar{A}\bar{B})C + ABD$
- $\textcolor{red}{F} = (\textcolor{red}{A} \odot \textcolor{red}{B})C + \textcolor{red}{A}B\textcolor{red}{D}$

Simplify the following expression and draw a logic diagram for the simplified expression

$$F = A B (D A + C \overline{D}) + \bar{A} \bar{B} (A + C)$$

